

Machine Learning for Improving Management of Acute and Chronic Pain

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Conflicts of Interest

- StatReviewer, Co-founder, Chief-scientist
- Statistical Editor/Consultant
 - Anesthesiology
 - Headache
- Funding: National Institute of Neurological Disorders and Stroke (NINDS)
 - RO1NS06525701
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Machine Learning

A computer program is said to learn from experience ...if its performance at tasks T , as measured by P , improves with experience E

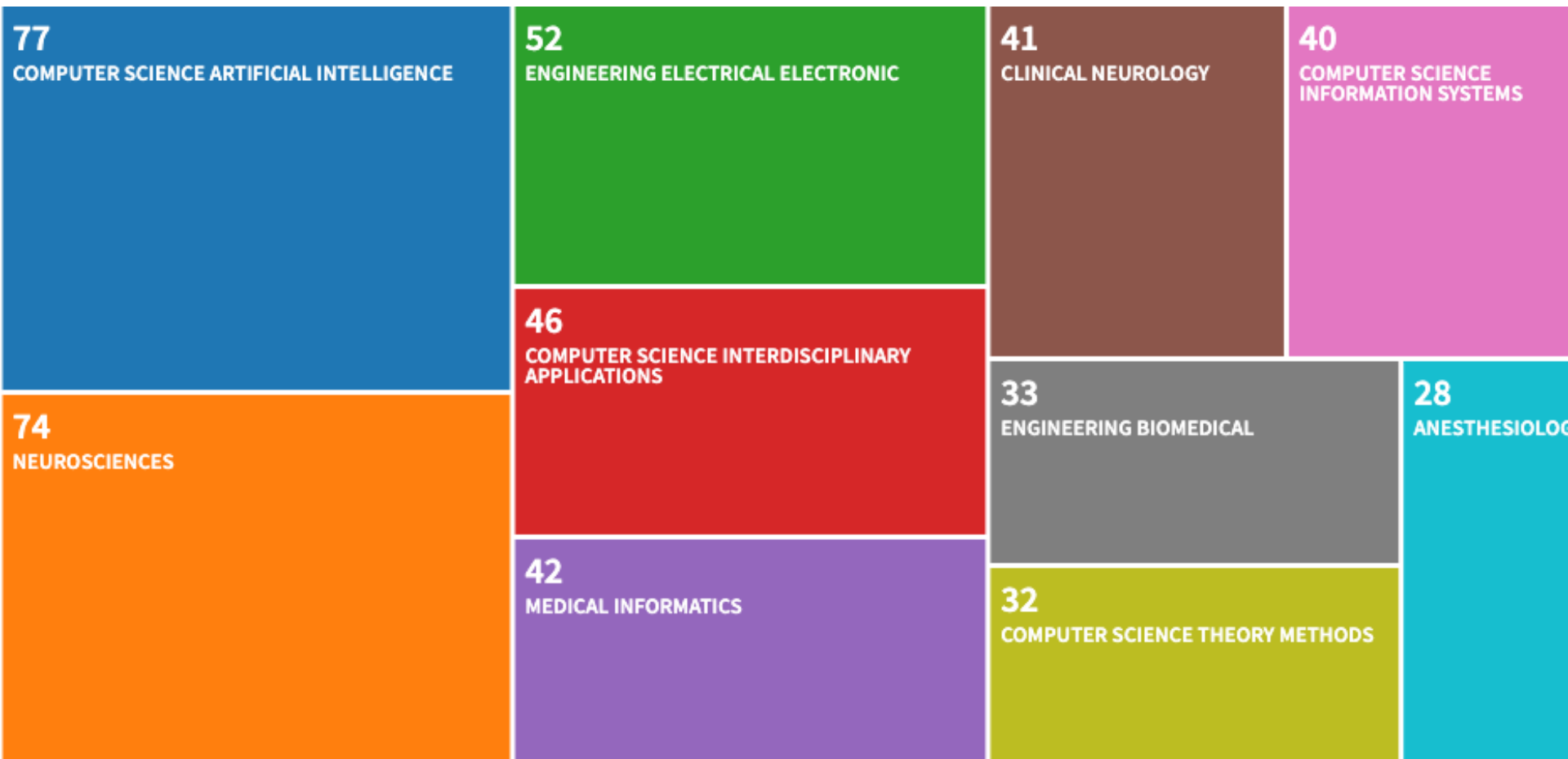
Machine Learning Types (Abridged)

Type	Description
Supervised Learning	Inputs, outputs both known, predict outputs
Unsupervised Learning	Inputs only, determine outputs
Reinforcement Learning	Given set of actions, maximize cumulative reward
Feature Learning	Given set of inputs, determine features
Sparse Dictionary Learning	Classify training data, determine if belongs to a previously learned class
Decision Tree Learning	Evaluate many inputs to predict class of output
Association Rule Learning	Identify, learn, and evolve 'rules' to characterize relationships among variables

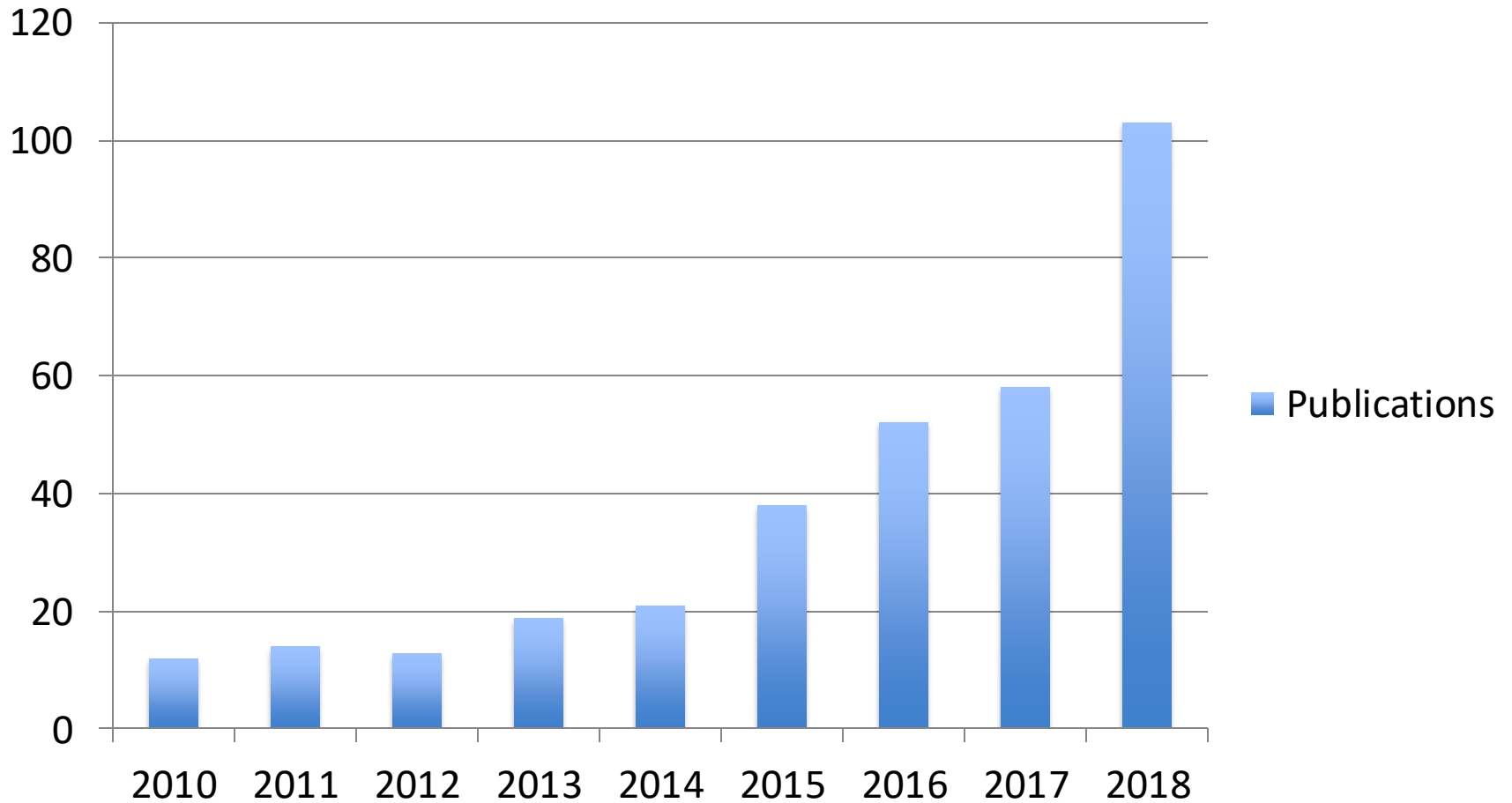
Common Machine Learning Models

- Artificial Neural Networks
 - Deep Learning
- Support Vector Machines
 - Classifier
- Bayesian (Neural) Networks
 - Probabilistic relationship inputs and outputs
- Genetic Algorithms
 - Search algorithms for finding solutions based on evolution of methods

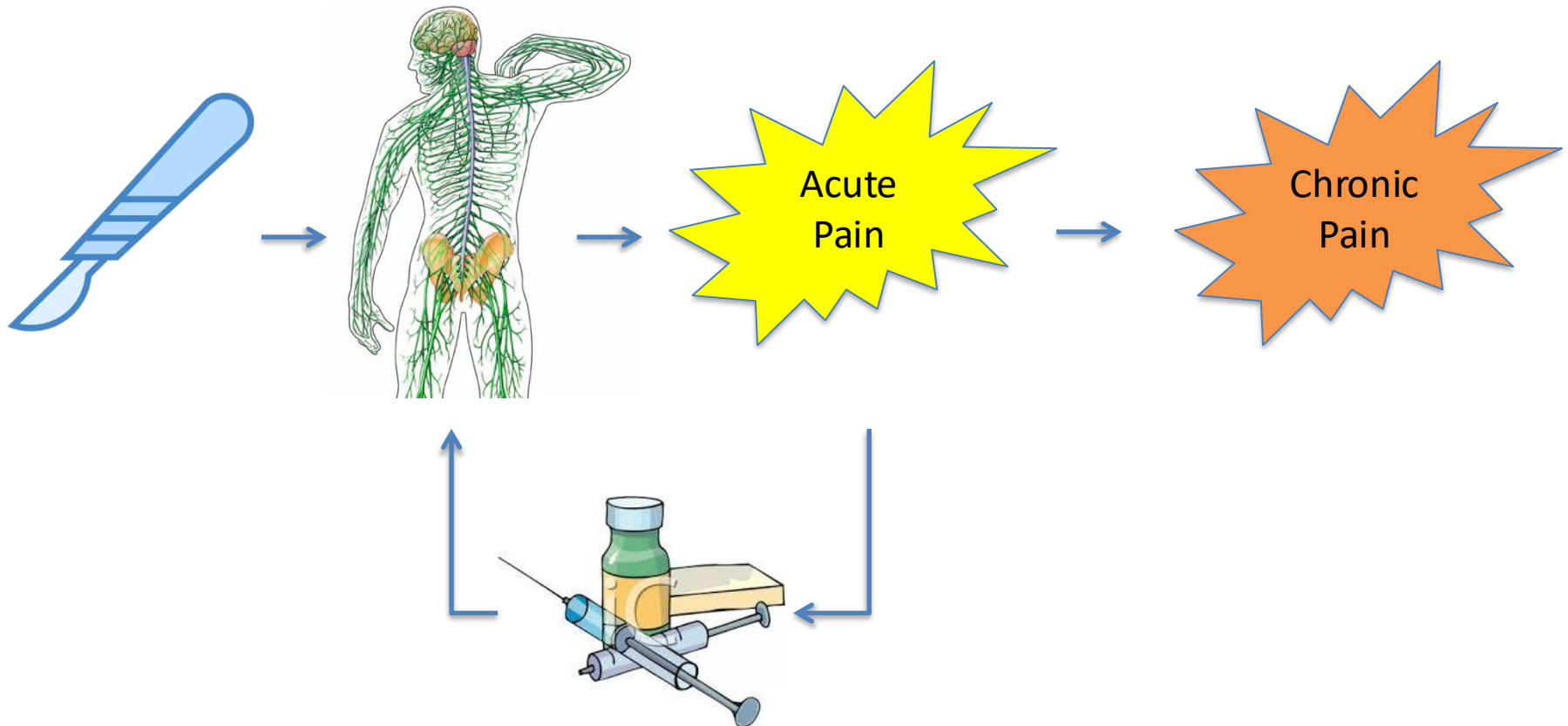
Applications in Pain



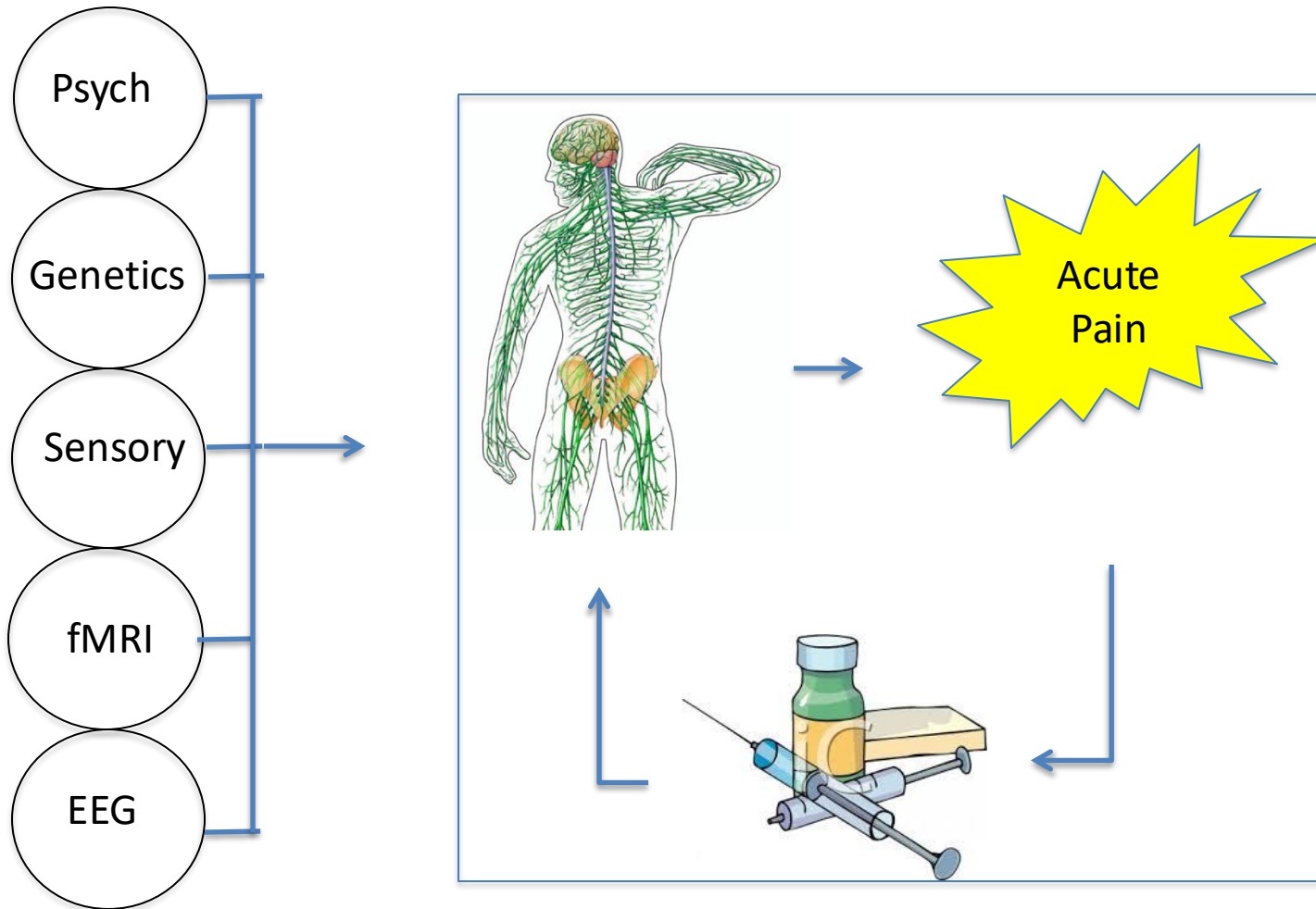
Machine Learning & Pain



Conceptual Model: Inputs to Outputs



High Dimensional Inputs



Teaching a Machine to Feel Postoperative Pain: Combining High-Dimensional Clinical Data with Machine Learning Algorithms to Forecast Acute Postoperative Pain

[Patrick J. Tighe, MD, MS](#) [Christopher A. Harle, PhD](#) [Robert W. Hurley, MD, PhD](#) [Haldun Aytug, PhD](#) [Andre P. Boezaart, MD, PhD](#) [Roger B. Fillingim, PhD](#)

- N = 8,071 surgical patients
- Predict moderate to severe pain Day 1-3 after surgery
 - Least Absolute Shrinkage and Selection Operator (LASSO)
 - gradient-boosted decision tree support vector machine (SVM)
 - neural network
 - k-nearest neighbor (k-NN)
 - logistic regression

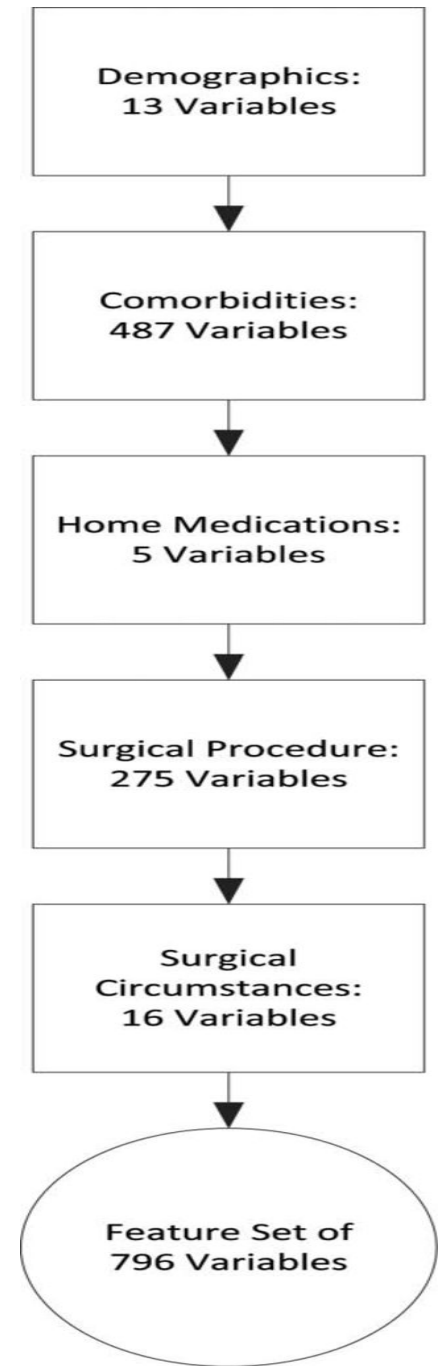
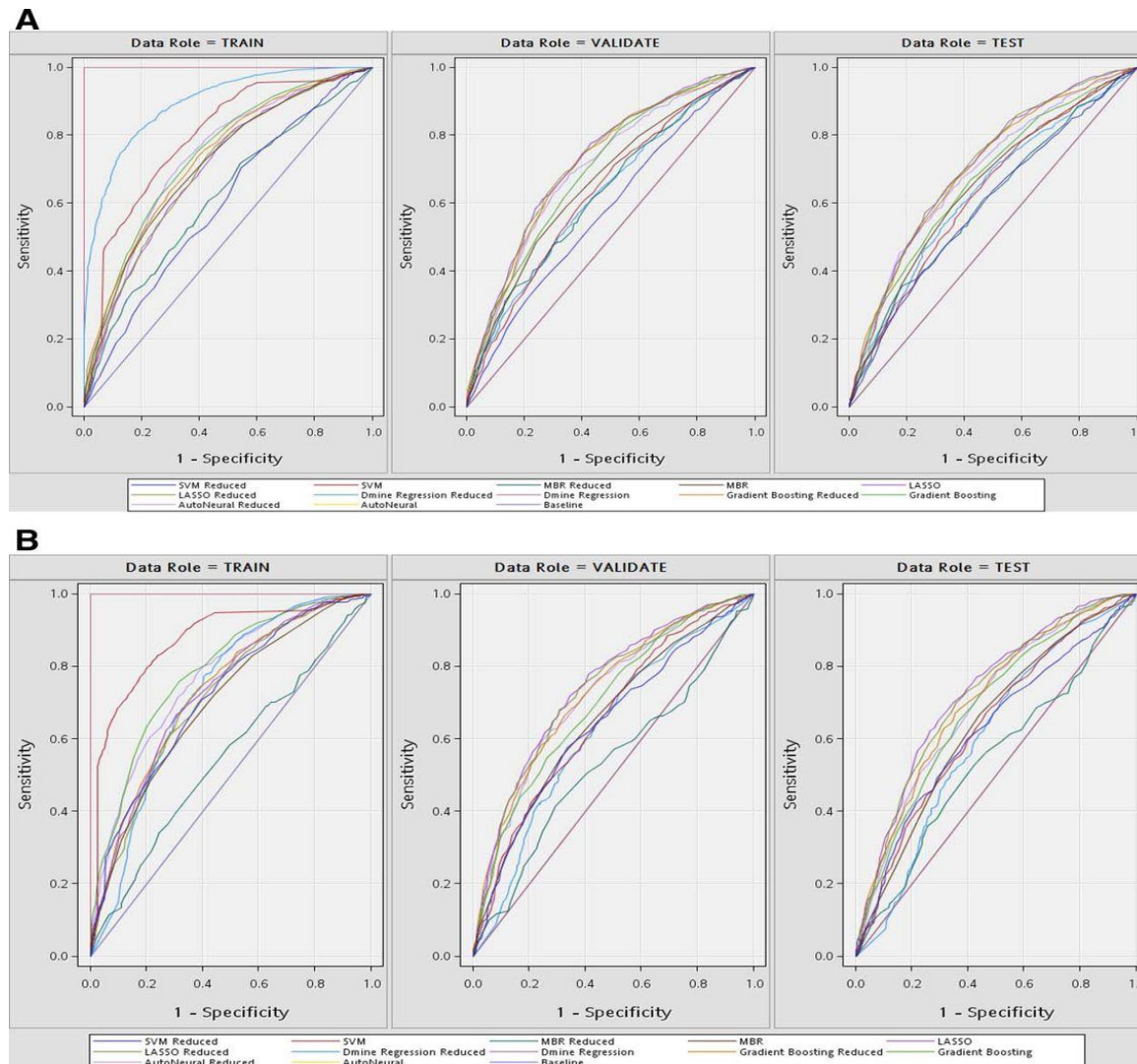


Figure 3 ROC for pain outcomes on POD 1 and 3. The ROC for each tested classifier are shown at the training, ...



Machine Learning in Pain

- Machine learning on encephalographic activity may predict opioid analgesia (Gram, 2015)
- An fMRI-based neurologic signature of physical pain (Wager, 2013)
- A data science approach to candidate gene selection of pain regarded as a process of learning and neural plasticity (Ultsch, 2016)

Ultsch et al. Pain. 2016 Dec 1;157(12):2747-57.

Gram et al. European Journal of Pain. 2015 Nov;19(10):1552-61.

Wager et al., New England Journal of Medicine. 2013 Apr 11;368(15):1388-97.

Other Inputs

- Facial recognition
 - Pain from changes in facial topography
- Radiology findings
 - Image processing
- Medical records
 - Natural language processing
- Microbiome

Future Challenges in Using Machine Learning to Improve Pain Management

- Machine Learning (ML) is *merely* numerical optimization
 - Optimism, over-fitting, variance-bias trade-off
- Many ML approaches do not properly consider uncertainty
 - Classification versus probabilistic prediction

Future Challenges in Using Machine Learning to Improve Pain Management

- Machine learning applications must consider calibration to be used in practice
 - Proper scoring rules (e.g., Brier Score)
- Classification/Decision must be considered in the context of cost (i.e., utility)
 - What is the cost of imprecision of pain/medication prediction?

Future Challenges in Using Machine Learning to Improve Pain Management

- Beware of over-hype!
 - Bayesian Neural Networks (Holy Roman Empire)
- ML should not allow us to avoid thinking about the problem

Thank You!

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